NOETHERIAN STEIN FACTORIZATION FOR SPACES: FINITE MIDDLE MAP, FIBRES KILLED BY THE THEOREM ON FORMAL FUNCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8126-8146 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8148-8194, the complete author proof (45 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (spaces, more general; both sides complete). The idempotent-lifting core is the same mechanism our wave-12 L16087-era thickening cards trade on.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemma stein-universally-closed (L16209); Cohomology of Spaces, lemmas proper-pushforward-coherent and formal-functions-stalk; Morphisms of Spaces permanence lemmas.

Source reference. Original theorem 03GY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7994–8014. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 035H: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8194–8194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 08B1: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8194–8194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 08AR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8194–8194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 03KR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8194–8194. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0462: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8194–8194. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1 (Stein factorization; Noetherian case)**.** *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of algebraic spaces over S with Y locally Noetherian. There exists a factorization*

$$\begin{array}{ccc} X & \xrightarrow{\quad} & Y' \\ & {\scriptstyle f'} \searrow \quad \swarrow {\scriptstyle \pi} & \\ & Y & \end{array}$$

with the following properties:

1. *the morphism f' is proper with connected geometric fibres,*
2. *the morphism $\pi : Y' \to Y$ is finite,*
3. *we have $f'_*\mathcal{O}_X = \mathcal{O}_{Y'}$,*
4. *we have $Y' = \underline{\mathrm{Spec}}_Y(f_*\mathcal{O}_X)$, and*
5. *Y' is the normalization of Y in X, see Morphisms, Definition 035H.*

Proof. Let $f = \pi \circ f'$ be the factorization of Lemma 03GY. Note that besides the conclusions of Lemma 03GY we also have that f' is separated (Morphisms of Spaces, Lemma 03KR) and finite type (Morphisms of Spaces, Lemma 0462). Hence f' is proper. By Cohomology of Spaces, Lemma 08AR we see that $f_*\mathcal{O}_X$ is a coherent $\mathcal{O}_Y$-module. Hence we see that π is finite, i.e., (2) holds.

This proves all but the most interesting assertion, namely that the geometric fibres of f' are connected. It is clear from the discussion above that we may replace Y by Y'. Then Y is locally Noetherian, $f : X \to Y$ is proper, and $f_*\mathcal{O}_X = \mathcal{O}_Y$. Let $\overline{y}$ be a geometric point of Y. At this point we apply the theorem on formal functions, more precisely Cohomology of Spaces, Lemma 08B1. It tells us that

$$\mathcal{O}^\wedge_{Y,\overline{y}} = \lim_n H^0(X_n, \mathcal{O}_{X_n})$$

where $X_n = \mathrm{Spec}(\mathcal{O}_{Y,\overline{y}}/\mathfrak{m}^n_{\overline{y}}) \times_Y X$. Note that $X_1 = X_{\overline{y}} \to X_n$ is a (finite order) thickening and hence the underlying topological space of X_n is equal to that of $X_{\overline{y}}$. Thus, if $X_{\overline{y}} = T_1 \amalg T_2$ is a disjoint union of nonempty open and closed subspaces, then similarly $X_n = T_{1,n} \amalg T_{2,n}$ for all n. And this in turn means $H^0(X_n, \mathcal{O}_{X_n})$ contains a nontrivial idempotent $e_{1,n}$, namely the function which is identically 1 on $T_{1,n}$ and identically 0 on $T_{2,n}$. It is clear that $e_{1,n+1}$ restricts to $e_{1,n}$ on X_n. Hence $e_1 = \lim e_{1,n}$ is a nontrivial idempotent of the limit. This contradicts the fact that $\mathcal{O}^\wedge_{Y,\overline{y}}$ is a local ring. Thus the assumption was wrong, i.e., $X_{\overline{y}}$ is connected as desired. □